ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1
PAPER 1 November 2019
2 hours 30 minutes
PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.
INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [ ].

1. Write 4000000 in standard form. **[2]**

2. Evaluate 8 + 3 × 3 − 6. **[2]**

3. In a class of Moleli, n(A) = 17 (play soccer), n(A ∩ B) = 5 (play both). How many play soccer only? **[3]**

4. Find the HCF and LCM of 18 and 16. **[4]**

5. Express 18 as a percentage of 40. **[3]**

6. Expand and simplify 2(x + 1) − 5x. **[3]**

7. A map of Mutare uses scale 2 : 800000. A road is 10 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm ↔ 1 km if 2:8 is first simplified — instead: scale 2 cm to 8 km.) **[4]**

8. Overnight in Mutare the temperature was 0°C. By 14:00 it was 27°C. Find the rise. **[3]**

9. A tank in Sakubva holds 40 litres. It is 2/4 full. How many litres are in the tank? **[3]**

10. Simplify 4^4 × 4^3. Leave as a power of 4. **[3]**

11. Simple interest on $200 at 10% per year for 2 years. Find the interest. **[4]**

12. Circumference of a circle radius 14 cm. Take $\pi = 22/7$. **[4]**

13. Work out (4 3 ; 1 0)(5 ; 2). **[4]**

14. Factorise $x^2 + 4x + 3$. **[4]**

15. Make h the subject of $A = 2\pi r(r + h)$. **[4]**

16. Solve<<
>>x + y = 11<<
>>x − y = 1. **[4]**

17. A cuboid water tank measures 12 m by 7 m by 7 m. Find its volume. **[3]**

18. A bag contains 6 red, 5 blue and 4 green beads. P(not green)? **[3]**

19. A sequence is 8, 11, 14, 17, … Find an expression for the nth term. **[4]**

20. A regular polygon has 6 sides. Find each exterior angle. **[3]**

21. In right-angled triangle XYZ, $\hat{X} = 60°$ and hypotenuse XZ = 12 cm. Find XY, adjacent to 60°. **[4]**

22. A triangular plot has base 6 m and perpendicular height 11 m. Find its area. **[3]**

23. Find the mean of 10, 4, 9, 8, 4. **[3]**

24. Solve $3(x + 1) = 12$. **[4]**

25. Find the midpoint of (4, 6) and (3, 3). **[3]**

26. Solve $4x + 1 < 25$. **[3]**

27. A ladder of length 20 m leans against a wall. The foot is 12 m from the wall. How far up the wall does it reach? **[4]**

28. Factorise completely $30x + 30$. **[3]**

29. In triangle ABC, $\hat{A} = 47°$ and $\hat{B} = 65°$. Find $\hat{C}$. **[3]**

30. Find the gradient of the line joining (4, 4) and (7, 10). **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).